

## NATIONAL UNIVERSITY OF SINGAPORE

SCHOOL OF COMPUTING  
FINAL ASSESSMENT FOR  
Semester 1 AY2025/26

## IT5008 Database Design and Programming

28 Nov 2025

Time Allowed 120 minutes

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**INSTRUCTIONS TO CANDIDATES**

1. This assessment paper contains **17** questions and comprises **8** printed pages, including this page.
2. More specific instructions *may* be given at the beginning of each section. Please read them carefully.
3. Write all your answers in the answer sheet provided. Please read the additional instruction for multiple choice (MCQ) and multiple response (MRQ) questions.
  - Any answer not in the space provided will not be graded.
  - For MCQ and MRQ, **shade your answer in the corresponding bubble on the answer sheet.**
  - For MCQ, if multiple answers are equally appropriate, pick one and shade **ONLY** the chosen answer on the answer sheet. Do **NOT** shade more than one answer.
  - For MRQ, shade **ALL** correct answers. Partial marks *may* be given, but not guaranteed.
  - Please erase the answers completely before shading new options.
4. Shade and write your student number in the answer sheet. Do **NOT** write your name.
5. The total marks for this assessment is **80 points**.
6. This is a **CLOSED-BOOK** assessment. You are only allowed to refer to one double-sided A4-size paper.
7. All SQL statements in this assessment paper are written for PostgreSQL 16.
8. We will use the monospace **NULL** to represent NULL-values. We will represent string with single-quote 'string'.
9. The last 2 page(s) is/are left blank. You may use them for your own work.
10. Submit only the answer sheet at the end of the assessment. You may keep the question paper.

| Question | 1 - 11 | 12 - 14 | 15 - 16 | 17 | Total     |
|----------|--------|---------|---------|----|-----------|
| Points   | 33     | 9       | 18      | 20 | <b>80</b> |

**GOOD LUCK!**

## Functional Dependencies and Normal Forms

For the next **eleven (11)** questions, we will be using the following relation  $R$  and set of functional dependencies  $\Sigma$ .

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, C\} \rightarrow \{B\}, \{B, C\} \rightarrow \{E\}, \{A, B\} \rightarrow \{C\}, \{C, D\} \rightarrow \{D, E\}, \{E\} \rightarrow \{B\} \}$$

**Notes:** We start with the following computation:

- **Keys:**  $\{ \{A, B, D\}, \{A, C, D\}, \{A, D, E\} \}$
- **Canonical Cover:**  $\{ \{A, B\} \rightarrow \{C\}, \{A, C\} \rightarrow \{B\}, \{B, C\} \rightarrow \{E\}, \{C, D\} \rightarrow \{E\}, \{E\} \rightarrow \{B\} \}$

1. (3 points) **(MRQ)** Select **ALL** the functional dependencies in  $\Sigma^+$  (i.e., logically entailed by  $\Sigma$ ).

- ☒  $\{A, E\} \rightarrow \{C\}$ 
☒  $\{A, B, D\} \rightarrow \{E\}$ 
☒  $\{A, B, C\} \rightarrow \{E\}$   
 B.  $\{A, E\} \rightarrow \{D\}$ 
 D.  $\{A, B, C\} \rightarrow \{D\}$

**Notes:** Compute the attribute closure of the left-hand side and check if the right-hand side is a subset.

2. (3 points) **(MRQ)** Select **ALL** the functional dependencies in the projection of  $\Sigma$  onto  $\{A, B, E\}$  (i.e.,  $\Sigma|_{\{A, B, E\}}$ ).

- A.  $\{A\} \rightarrow \{E\}$ 
☒  $\{A, E\} \rightarrow \{B\}$ 
 E.  $\{B, E\} \rightarrow \{A\}$   
☒  $\{E\} \rightarrow \{B\}$ 
☒  $\{A, B\} \rightarrow \{E\}$

**Notes:**  $\{A, E\} \rightarrow \{B\}$  is logically implied. Can be derived since  $\{E\} \rightarrow \{B\}$  is in the projection.

3. (3 points) **(MRQ)** Select **ALL** the *superkeys* of  $R$  with  $\Sigma$  from the choice below.

- ☒  $\{A, B, C, D\}$ 
☒  $\{A, B, D, E\}$ 
 E.  $\{B, C, D, E\}$   
 B.  $\{A, B, C, E\}$ 
☒  $\{A, C, D, E\}$

**Notes:** Find all superset of keys.

4. (3 points) **(MRQ)** Select **ALL** the *candidate keys* (i.e., keys) of  $R$  with  $\Sigma$  from the choice below.

- A.  $\{A, B, C\}$ 
 C.  $\{A, B, E\}$ 
☒  $\{A, D, E\}$   
☒  $\{A, B, D\}$ 
☒  $\{A, C, D\}$

**Notes:** We should always compute keys first.

5. (3 points) **(MRQ)** Select **ALL** the *prime attributes* of  $R$  with  $\Sigma$ .

- ☒  $A$ 
☒  $B$ 
☒  $C$ 
☒  $D$ 
☒  $E$

**Notes:** Union of all candidate keys.

6. (3 points) **(MRQ)** We say that two sets of attributes  $X$  and  $Y$  are *functionally equivalent* if  $X \rightarrow Y$  and  $Y \rightarrow X$ . Select **ALL** the sets of attributes that are functionally equivalent to  $\{E\}$ .

A.  $\{A, C\}$ C.  $\{B, C\}$ E.  $\{B, C, D\}$ B.  $\{A, D\}$ D.  $\{C, D\}$ 

**Notes:** The attribute closure of  $\{E\}$  is  $\{E\}^+ = \{B, E\}$ . So the only candidate must involve only  $\{B\}$ . Hence, there is no answer. Just leave this blank.

7. (3 points) **(MRQ)** Select **ALL** the functional dependencies below that violates the BCNF property of  $R$  with  $\Sigma$ .

✓  $\{E\} \rightarrow \{B\}$ ✓  $\{A, C\} \rightarrow \{B\}$ 

E. None of the above.

✓  $\{A, B\} \rightarrow \{C\}$ ✓  $\{B, C\} \rightarrow \{E\}$ 

**Notes:** Check that (i) LHS is not a superset of any candidate key.

8. (3 points) **(MRQ)** Select **ALL** the functional dependencies below that violates the 3NF property of  $R$  with  $\Sigma$ .

A.  $\{E\} \rightarrow \{B\}$ C.  $\{A, C\} \rightarrow \{B\}$ 

✓ None of the above.

B.  $\{A, B\} \rightarrow \{C\}$ D.  $\{B, C\} \rightarrow \{E\}$ 

**Notes:** Check that (i) LHS is not a superset of any candidate key and (ii) RHS is not a prime attribute.

9. (3 points) **(MCQ)** Consider the following decomposition of  $R$  with  $\Sigma$ .

(i).  $\{\{A, B, C\}, \{A, C, D\}, \{B, E\}\}$ (ii).  $\{\{A, C, E\}, \{A, B, D\}, \{B, E\}\}$ (iii).  $\{\{A, C, E\}, \{A, C, D\}, \{B, E\}\}$ 

Which decompositions from the previous page are lossless-join decomposition of  $R$  with  $\Sigma$  in BCNF? Select the most appropriate option.

A. (i) only

C. (i) and (ii) only

E. (i), (ii), and (iii)

B. (iii) only

✓ (ii) and (iii) only

**Notes:** Find a path of binary lossless-join decomposition. One path for (ii) and (iii) is to notice  $\{E\} \rightarrow \{B\}$ , so we need to find another fragment with a common attribute  $\{E\}$ .

For (i),  $\{E\}$  is only available in  $\{B, E\}$ . Since there is no other functional dependencies such that the left-hand side is a single attribute (you can check from the minimal cover), there is no way this can be lossless-join.

10. (3 points) **(MCQ)** Consider the following set of functional dependencies.

(i).  $\{\{A, B\} \rightarrow \{C\}, \{A, C\} \rightarrow \{B\}, \{B, C\} \rightarrow \{E\}, \{C, D\} \rightarrow \{E\}, \{E\} \rightarrow \{B\}\}$ (ii).  $\{\{A, B\} \rightarrow \{C\}, \{A, C\} \rightarrow \{E\}, \{B, C\} \rightarrow \{E\}, \{C, D\} \rightarrow \{E\}, \{E\} \rightarrow \{B\}\}$ (iii).  $\{\{A, B\} \rightarrow \{C\}, \{A, C\} \rightarrow \{B\}, \{B, C\} \rightarrow \{E\}, \{C, D\} \rightarrow \{B\}, \{E\} \rightarrow \{B\}\}$ 

Which set of functional dependencies above are the minimal cover of  $\Sigma$ ? Select the most appropriate option.

- A. (i) only  
 B. (iii) only  
 C. (i) and (ii) only  
 D. (ii) and (iii) only  
 ✓ (i), (ii), and (iii)

**Notes:** One way is to check that these are equivalent to  $\Sigma$ . Then run minimal cover algorithm to check if it can be simplified further. If not, then it is a minimal cover.

11. (3 points) **(MCQ)** Consider the following decomposition of  $R$  with  $\Sigma$ .

$$\{ \{A, D, E\}, \{A, C, E\}, \{B, E\} \}$$

Select the most appropriate statement about the decomposition.

- A. The decomposition is not a valid decomposition (i.e., it loses some attributes).  
 B. The decomposition is not a lossless-join decomposition.  
 ✓ The decomposition is not a dependency preserving decomposition.  
 D. There are some fragments in the decomposition that is not in BCNF.  
 E. There are some fragments in the decomposition that is not in 3NF.

**Notes:** For lossless-join we use the similar trick to first combine  $\{B, E\}$  as it can be trivially losslessly decomposed as long as the other fragment has  $\{E\}$ . From there, we can check  $\{A, E\}^+$  and see if either  $C$  or  $D$  is inside.

For dependency preserving decomposition, we compute the projection.

- $\Sigma|_{\{A,D,E\}} = \emptyset$
- $\Sigma|_{\{A,C,E\}} = \{ \{A, C\} \rightarrow \{E\}, \{A, E\} \rightarrow \{C\} \}$
- $\Sigma|_{\{B,E\}} = \{ \{E\} \rightarrow \{B\} \}$

By inspection,  $\{B, C\} \rightarrow \{E\}$  is not preserved. Also, we can see that all fragments are in BCNF. They are also automatically in 3NF.

*Continue on the next page...*

## Relational Algebra

For the next **three (3)** questions, we will be using the following schema.

- Customer(cid, name) with (cid) as the primary key.
- Order(oid, total, date, cid) with (oid) as the primary key and Order.cid referencing Customer.cid.
- Product(pid, name, price) with (pid) as the primary key.
- Includes(oid, pid, qty) with (oid, pid) as the primary key, Includes.oid referencing Order.oid, and Includes.pid referencing Product.pid.
- Rates(cid, pid, score) with (cid, pid) as the primary key, Rates.cid referencing Customer.cid, and Rates.pid referencing Product.pid.

12. (3 points) **(MCQ)** Consider the following relational algebra expression.

- (i).  $\pi_{\text{name}}(\sigma_{\text{total} > 1000}(\text{Customer} \bowtie \text{Order}))$
- (ii).  $\pi_{\text{name}}(\text{Customer} \bowtie \sigma_{\text{total} > 1000}(\text{Order}))$
- (iii).  $\pi_{\text{name}}(\sigma_{\text{cid} > 1000}(\text{Customer}) \bowtie \text{Order})$

Which relational algebra expression above finds the different names of customers that made at least one order worth more than 1000 in total?

- |               |                        |                         |
|---------------|------------------------|-------------------------|
| A. (i) only   | ✓ (i) and (ii) only    | E. (i), (ii), and (iii) |
| B. (iii) only | D. (ii) and (iii) only |                         |

**Notes:** cid > 1000 does not make sense.

13. (3 points) **(MCQ)** Consider the following relational algebra expression.

- (i).  $\pi_{P_1.\text{name}}(\rho(\text{Product}, P_1) \bowtie (\pi_{P_2.\text{pid}}(\rho(\text{Product}, P_2)) - \pi_{I.\text{pid}}(\rho(\text{Includes}, I))))$
- (ii).  $\pi_{P_1.\text{name}}(\rho(\text{Product}, P_1) \bowtie (\rho(\text{Product}, P_2) - \rho(\text{Includes}, I)))$
- (iii).  $\pi_{P_1.\text{name}}(\rho(\text{Product}, P_1) \bowtie \pi_{P_2.\text{pid}}(\rho(\text{Product}, P_2) - \rho(\text{Includes}, I)))$

Which relational algebra expression above finds the different name of products that have never been purchased?

- |               |                        |                         |
|---------------|------------------------|-------------------------|
| ✓ (i) only    | C. (i) and (ii) only   | E. (i), (ii), and (iii) |
| B. (iii) only | D. (ii) and (iii) only |                         |

**Notes:** In (ii) and (iii),  $\rho(\text{Product}, P_2) - \rho(\text{Includes}, I)$  is invalid as it is not union compatible.

14. (3 points) **(MCQ)** Consider the following relational algebra expression.

- (i).  $\pi_{\text{name}}(\sigma_{\text{cid}=1}(\pi_{\text{cid}}(\text{Customer})) \bowtie (\sigma_{\text{score} \geq 3.0}(\text{Rates})) \bowtie \text{Products})$
- (ii).  $\pi_{\text{name}}(\sigma_{[(\text{cid}=1) \wedge (\text{score} \geq 3.0)]}(\pi_{\text{cid}}(\text{Customer})) \bowtie (\text{Rates}) \bowtie \text{Products})$
- (iii).  $\pi_{\text{name}}(\sigma_{\text{score} \geq 3.0}(\pi_{\text{cid}}(\text{Customer})) \bowtie (\sigma_{\text{cid}=1}(\text{Rates})) \bowtie \text{Products})$

Which relational algebra expression above finds the different name of products that the customer with cid = 1 has rated 3.0 or higher?

- ✓ (i) only  
B. (iii) only

- C. (i) and (ii) only  
D. (ii) and (iii) only

- E. (i), (ii), and (iii)

**Notes:** Under normal situation, the conditions can be moved around. But in this case, the condition for  $\sigma$  need to use the attribute available in the relation. The issue is that `score` is not available in `Customer`. Hence, (ii) and (iii) are actually invalid.

## Cardinalities

15. (10 points) **(FITB)** Consider two arbitrary relations  $R$  and  $S$  with the following properties.

- $R$  has  $m$  rows and  $S$  has  $n$  rows such that  $m > n > 0$ .
- $R$  and  $S$  are union-compatible.
- $R$  and  $S$  contains no null values.
- There are no other assumptions on  $R$  and  $S$ .

For each relational algebra expressions below, find the maximum and minimum number of rows in the resulting relation. Specify your answer as the following possible values.

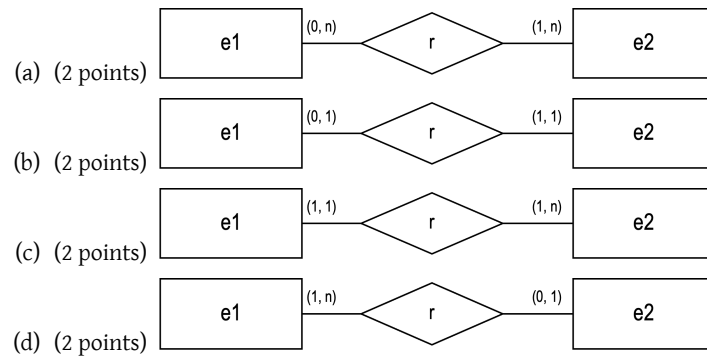
- **Constants:** 0, 1, ...
- **Variables:**  $m$  or  $n$ .
- **Binary Operation:**  $m + n$ ,  $m - n$ ,  $m \times n$ , or  $m/n$ .

- (a) (2 points)  $R \cup S$  (i.e., union).  
 (b) (2 points)  $R \cap S$  (i.e., intersection).  
 (c) (2 points)  $R - S$  (i.e., set difference).  
 (d) (2 points)  $R \bowtie S$  (i.e., natural join).  
 (e) (2 points)  $R \Join S$  (i.e., natural full outer join).

**Notes:**

| Expression    | Minimum                                                          | Maximum                                           |
|---------------|------------------------------------------------------------------|---------------------------------------------------|
| $R \cup S$    | $m$<br>(when $S \subset R$ )                                     | $m + n$<br>(when $S \cap R = \emptyset$ )         |
| $R \cap S$    | 0<br>(when $S \cap R = \emptyset$ )                              | $n$<br>(when $S \subset R$ )                      |
| $R - S$       | $m - n$<br>(when $S \subset R$ )                                 | $m$<br>(when $S \cap R = \emptyset$ )             |
| $R \bowtie S$ | 0<br>(when $R$ and $S$ has common attr, but not equal)           | $m \times n$<br>(when $S$ and $R$ no common attr) |
| $R \Join S$   | $m$<br>(when $S$ has no dangling and $R$ match to only one $S$ ) | $m \times n$<br>(when $S$ and $R$ no common attr) |

16. (8 points) **(FITB)** For each entity-relationship diagram below we omit the attributes as they are not relevant. Consider a case where there are 100 entities for  $e_2$ . Find the maximum and minimum number of entities for  $e_1$ . If there are no known upper bound, write  $\infty$  as the answer.

**Notes:**

| ERD | Minimum                             | Maximum                          |
|-----|-------------------------------------|----------------------------------|
| (a) | 1<br>(at least one $e_1$ is needed) | $\infty$<br>(unbounded)          |
| (b) | 100<br>(1-to-1 matching)            | $\infty$<br>(unbounded)          |
| (c) | 100<br>(1-to-1 matching)            | $\infty$<br>(unbounded)          |
| (d) | 0<br>(unmatched $e_2$ )             | 100<br>(fully matched to $e_2$ ) |

*Continue on the next page...*

## Miscellaneous

17. (20 points) Consider the following relation and set of functional dependencies. Note that the set of functional dependencies is already given in canonical cover.

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A\} \rightarrow \{C\}, \{B\} \rightarrow \{A, C\}, \{C\} \rightarrow \{A, B\}, \{D, E\} \rightarrow \{A\}, \{A\} \rightarrow \{E\} \}$$

- (a) (4 points) Find all the *candidate keys* (i.e., keys) of  $R$  with  $\Sigma$ . (**Note:** You do not have to show your work).

**Notes:**  $\{ \{A, D\}, \{B, D\}, \{C, D\}, \{D, E\} \}$

- (b) (5 points) Find one lossless-join decomposition of  $R$  with  $\Sigma$  in BCNF using the algorithm introduced in the course.

**Notes:** One possible step is as follows.

- Using  $\{A\} \rightarrow \{C\}$  as a violation since  $\{A\}$  is not a superkey.
- Compute  $\{A\}^+ = \{A, B, C, E\}$ .
- Decompose  $R = \{A, B, C, D, E\}$  into the following fragments.
  - $R_1 = \{A, B, C, E\}$ 
    - \*  $\Sigma|_{R_1} = \{ \{A\} \rightarrow \{A, B, C, E\}, \{B\} \rightarrow \{A, B, C, E\}, \{C\} \rightarrow \{A, B, C, E\} \}$ .
    - \*  $R_1$  is in BCNF.
  - $R_2 = \{A, D\}$ .
    - \*  $\Sigma|_{R_2} = \emptyset$ .
    - \*  $R_2$  is in BCNF.

One possible answer is:  $\{ \{A, B, C, E\}, \{A, D\} \}$ .

- (c) (2 points) Is your BCNF decomposition above a dependency preserving decomposition? Simply answer YES or NO.

**Notes:** NO. For the answer above  $\{D, E\} \rightarrow \{A\}$  is not preserved.

- (d) (5 points) Find one lossless-join dependency preserving decomposition of  $R$  with  $\Sigma$  in 3NF using the algorithm introduced in the course.

**Notes:** Starting from our  $\Sigma$ , there are two likely minimal cover you will find. Other minimal cover can be obtained from  $\Sigma^+$ .

- $\{ \{A\} \rightarrow \{C\}, \{A\} \rightarrow \{E\}, \{B\} \rightarrow \{A\}, \{C\} \rightarrow \{B\}, \{D, E\} \rightarrow \{A\} \}$ 
  - Canonical:  $\{ \{A\} \rightarrow \{C, E\}, \{B\} \rightarrow \{A\}, \{C\} \rightarrow \{B\}, \{D, E\} \rightarrow \{A\} \}$
  - Construction:  $\{ \{A, C, E\}, \{A, B\}, \{B, C\}, \{A, D, E\} \}$
  - No subsumption needed and key is already subsumed.
- $\{ \{A\} \rightarrow \{C\}, \{A\} \rightarrow \{E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{A\}, \{C\} \rightarrow \{B\}, \{D, E\} \rightarrow \{A\} \}$ 
  - Canonical:  $\{ \{A\} \rightarrow \{C, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{A, B\}, \{D, E\} \rightarrow \{A\} \}$
  - Construction:  $\{ \{A, C, E\}, \{B, C\}, \{A, B, C\}, \{A, D, E\} \}$
  - Subsumption:  $\{ \{A, C, E\}, \{A, B, C\}, \{A, D, E\} \}$
  - Key is already subsumed.

- (e) (4 points) A naïve approach from part (d) will not produce the smallest number of fragments. We can find a decomposition with fewer fragments that satisfies the following criteria.

- The decomposition is both lossless-join and dependency preserving.
- The number of fragment is exactly 2 and all fragments are in 3NF.



Find one such decomposition.

**Notes:**  $\{\{A, D, E\}, \{A, B, C, E\}\}$

We can show that this satisfies all the properties. Let  $R_1 = \{A, D, E\}$  and  $R_2 = \{A, B, C, E\}$ .

- Lossless-Join
  - Common attributes:  $\{A, D, E\} \cap \{A, B, C, E\} = \{A, E\}$ .
  - Attribute closure:  $\{A, E\}^+ = \{A, B, C, E\} \supseteq R_2$ .
- Dependency Preserving
  - $\Sigma|_{R_1} = \{\{A\} \rightarrow \{E\}, \{D, E\} \rightarrow \{A\}\}$ .
  - $\Sigma|_{R_2} = \{\{A\} \rightarrow \{A, B, C, E\}, \{B\} \rightarrow \{A, B, C, E\}, \{C\} \rightarrow \{A, B, C, E\}\}$ .
  - $\Sigma_{\cup} = \{\{A\} \rightarrow \{E\}, \{D, E\} \rightarrow \{A\}, \{A\} \rightarrow \{A, B, C, E\}, \{B\} \rightarrow \{A, B, C, E\}, \{C\} \rightarrow \{A, B, C, E\}\}$ .
  - By inspection, this logically entails all functional dependencies in  $\Sigma$ .
- 3NF
  - $\Sigma|_{R_1} = \{\{A\} \rightarrow \{E\}, \{D, E\} \rightarrow \{A\}\}$ .
    - \* Candidate keys:  $\{\{A, D\}, \{D, E\}\}$ .
    - \* Since the prime attributes contain all attributes, this is trivially in 3NF.
  - $\Sigma|_{R_2} = \{\{A\} \rightarrow \{A, B, C, E\}, \{B\} \rightarrow \{A, B, C, E\}, \{C\} \rightarrow \{A, B, C, E\}\}$ .
    - \* All left-hand side in this functional dependencies are superkeys.
    - \* Hence, it is not only in 3NF, it is in BCNF.

**Notes:** The working required need not be this detailed. We show you the complete working so that you can check your work.

Also, Part (d) and Part (e) are voided because of the following reasons.

- $\Sigma$  is not in canonical unlike what the claim says.
- $R$  is already in 3NF with respect to  $\Sigma$ . So the best decomposition is basically  $\{\{A, B, C, D, E\}\}$ . In other words, do not even split.

END OF PAPER